

# KHALIL AHMAD QAMAR

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## TECHNICAL SKILLS

**Languages & Systems:** Python, C/C++, SQL, Bash, Linux, Git, Docker, AWS, GCP, Azure, PostgreSQL, FastAPI, Flask, XML

**Libraries & Tools:** PyTorch, scikit-learn, NumPy, SciPy, Pandas, xarray, OpenCV, Matplotlib, pytest, LangChain, Pinecone, HDF5

## WORK EXPERIENCE

### MLOps Engineering Intern | Palitronica Inc.

May 2026 – August 2026

- Built and maintained core Python ML infrastructure for RF-signal anomaly detection, shipping 10+ capabilities across new models, multidimensional inference, HDF5 persistence, deterministic hyperparameter search, and OOM/retry recovery.
- Built KDE/covariance anomaly models and production ML lifecycle infrastructure, wiring Azure Cosmos DB and HDF5 model artifacts across training, retraining, and versioned publishing to enable reproducible inference without retraining.
- Root-caused and fixed multidimensional xarray/NumPy inference failures in Dockerized ML pipelines on Linux nodes, preserving anomaly labels, scores, and coordinate integrity across split/merge execution with pytest regression tests.
- Built OOM/retry recovery and deterministic Cartesian hyperparameter search for resilient, reproducible model tuning.
- Designed a human-in-the-loop agentic AI workflow with custom Copilot agents/skills and context engineering, turning tickets and repository context into scoped execution plans with validation gates to accelerate engineering throughput.

### Computer Vision Research Intern | Unity Health Toronto

August 2025 – December 2025

- Fine-tuned a SAM-based model for ICU diaphragm ultrasound segmentation by curating COCO datasets from raw frames and ground-truth clinician masks, converting masks into point/box prompts in Python, improving segmentation by 70%.
- Automated segmentation evals with PyTorch/MMDetection/MMEEngine across IoU, Dice, and mAP, saving 15+ hours/week.
- Built NumPy/Pandas/Matplotlib utilities to process RF ultrasound signals, enhance frames, and visualize them in a GUI.

### Software Engineer Intern | Untether AI

January 2025 – April 2025

- Built a Python/Pandas GUI ETL pipeline for 2,000+ AI chip logs across 10+ formats, automating 40+ hours/week of work.
- Drove chip-yield analytics for 200+ test types and XML schema validation for 100+ definitions, catching malformed data.

## STUDENT TEAMS AND RESEARCH EXPERIENCE

### Machine Learning Engineer | WAT.ai - InsightPulse | GitHub

January 2026 – Present

- Built a financial scenario engine forecasting 21-day moves across 12 market/sector targets from 8 macro shocks, using 58 event-aware features to quantify baseline, hypothetical, and incremental market impacts.
- Trained 12 scikit-learn Ridge models on time-aligned macro/market data, preventing look-ahead leakage in forecasts.
- Built FastAPI/PostgreSQL services and LLM-generated market-state summaries with evals for reliable scenario outputs.

### Geospatial AI Research Assistant | Vision and Image Processing Lab | Notebook

January 2026 – Present

- Built a Python/GEE pipeline pairing AlphaEarth embeddings with ESA WorldCover labels across 3 regions (13K samples/region) to test whether pretrained satellite embeddings encode recoverable land-cover semantics.
- Built a KMeans urban/non-urban clustering baseline, achieving 86.2–96.8% accuracy against labels across 3 regions.

## SELECTED PROJECTS

### PulseAI | GitHub · Process | Python, Flask, AWS, Docker, LangChain, Pinecone

December 2025 – Present

- Built a RAG medical chatbot over clinical PDFs using Pinecone semantic retrieval for grounded LLM responses.
- Built PDF ingestion, chunking, and embedding pipelines; deployed with Flask, Docker, AWS EC2, and GitHub Actions CI/CD.

### AI Voice Assistant | GitHub · Demo | Python, LiveKit, FastAPI, OpenAI, ElevenLabs, SQLite

May 2025 – June 2025

- Built a real-time conversational AI assistant using LiveKit for low-latency and OpenAI + ElevenLabs for duplex speech.
- Designed a multi-user backend with Flask/FastAPI and SQLite persistence, integrating LLM-driven intent handling.

## EDUCATION

### University of Waterloo

September 2023 – Present

BASc, Mechatronics Engineering Co-op, Artificial Intelligence Minor | GPA: 3.7

Waterloo, ON